Can VIX Improve Intraday Trend Following?

How to use option-implied volatility to better calibrate intraday breakout bands and position sizing.

CONCRETUM RESEARCH
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Over the past few years, we have published a series of research articles documenting the robustness of **intraday trend-following models on SPY** and other liquid equity index products.

The appeal of these strategies is not only their standalone return profile. In many cases, their real value comes from the fact that, by construction, they **deliberately take orthogonal exposure from a passive long-only equity allocation**. This can translate into a **significant benefit in terms of correlation control and alpha generation.**

Our public work on this topic started with **Beat the Market: An Effective Intraday Momentum Strategy for S&P500 ETF (SPY)**, published in 2024, where we showed how **intraday demand and supply imbalances** can be transformed into a **systematic trend-following** framework.

Most of the models we have presented so far share a common design choice: they use recent **realized volatility to define the intraday breakout levels** and, in many cases, **to size positions dynamically**.

That choice is simple, transparent, and hard to dislike. But it is also **backward-looking**.

This naturally leads to a question we wanted to test more carefully:

> **Can implied volatility improve the precision with which we define breakout levels and size intraday trades?**

In this article, we introduce the **full rules** behind an **effective intraday volatility breakout strategy on SPY** that compounds at 17% per year (net of cost) over the backtest window, while showing **an annualized alpha of ~18%** and a **beta** of **-0.06 versus SPY**.

Performance Statistics

Metric	Value
CAGR	17.22%
Alpha (Yearly)	17.89%
Beta vs SPY	-0.06
Correlation vs SPY	-0.08

We then replace historical volatility with **VIX** readings when **calibrating the bands** and assess whether option-implied volatility can improve the effectiveness of the strategy. We will show that **VIX** data can be used **more effectively** when we **avoid** introducing **unnecessary lag** in the volatility input.

Moreover, we ask whether implied volatility can be used for another part of the model: **position sizing**.

Finally, we assess whether VIX information becomes more useful during macro-announcement days such as FOMC, CPI, ISM, Initial Jobless Claims, and Nonfarm Payrolls.

There are several useful lessons in this article. Enjoy your read.

Base Strategy

Our base model is a simple intraday breakout strategy applied to SPY.

For each trading day, we define an upper and lower band around the relevant opening reference. The width of the band is set using the prior 10-day historical volatility multiplied by a volatility multiplier of 0.4.

Let O_t be today's open and C_{t-1} be yesterday's close. To account for overnight gaps, the upper and lower reference prices are defined as:

$$P_t^+ = \max(O_t, C_{t-1}), \qquad P_t^- = \min(O_t, C_{t-1})$$

Given a volatility input $sigma_t$ used for the bands and a multiplier *VM*, the breakout boundaries are:

$$UB_t = P_t^+ \times (1 + VM \cdot \sigma_t^B)$$

$$LB_t = P_t^- \times (1 - VM \cdot \sigma_t^B)$$

The strategy can enter after the first 30 minutes of the session (i.e. 10am ET) and checks signals every 30 minutes. A long position is opened when price breaks above the upper band, while a short position is opened when price breaks below the lower band.

Positions are closed when price moves back inside the noise area (the area between the two bands), or at the end of the trading day.

Position sizing follows a standard volatility-targeting rule.

The number of SPY shares to trade on day t is equal to:

$$Shares_t = \min\left(4, \frac{2\%}{\sigma_t^S}\right) \times \frac{AUM_{t-1}}{Open_t}$$

where 2% is the daily volatility target, while 4 represents the maximum leverage allowed in the portfolio.

> ***Backtest assumptions**. The backtest is conducted from January 4, 2007 until June 30, 2026. Results are intentionally gross of transaction costs because the objective of this article is to compare two different ways of calibrating and sizing the same intraday trend framework.*

Why Should We Use VIX?

Historical volatility is a useful proxy for future volatility, but it is blind to upcoming scheduled events.

If the next session contains **CPI, Nonfarm Payrolls, an FOMC-related release**, or another event that investors already know may move the market, a **backward-looking volatility measure may underestimate the range** that traders are actually preparing for.

This is where implied volatility becomes appealing.

The **VIX Index**, calculated by Cboe from SPX option prices, is designed to represent a **constant-maturity 30-day expectation of S&P 500 volatility**. In other words, it embeds the price investors are willing to pay for future uncertainty, rather than simply extrapolating what happened over the past few sessions.

There is, however, an immediate **problem**: **VIX is not an unbiased estimate of future realized volatility.**

For more details about the structural overpricing of implied volatility and how investors try to exploit it, you can read our previous article

Automating a Volatility Strategy With Python and Interactive Brokers

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Because investors tend to pay a premium for downside protection, **implied volatility is usually above subsequent realized volatility.** This well-known volatility risk premium can severely impact our tests. In fact, if we simply use raw VIX data divided by the square root of 252 to set intraday breakout bands, the bands will generally sit farther away from the open than the historical-volatility bands.

That would make the comparison unfair. We would not be testing whether VIX contains better information. We would mostly be testing a strategy with wider bands.

Let's provide a simple numerical example to clarify this issue. Over the sample used in this study, the average raw daily VIX estimate is roughly 25% higher than the average historical-volatility estimate.

To keep the comparison cleaner, we therefore adjust VIX by a constant scale factor so that its average daily volatility level matches the average historical volatility level over the overlapping sample.

This produces a **scale factor of 0.74.**

We thus define the adjusted implied-volatility (or *adjusted VIX*) input as:

$$\sigma_t^{VIX,adj} = \sigma_t^{VIX,raw} \times \frac{\mathrm{mean}(\sigma^{Hist})}{\mathrm{mean}(\sigma^{VIX,raw})}$$

The adjustment does not make the VIX model equal to the historical-volatility model. It only removes the structural bias. Day by day, the two measures can still diverge meaningfully, which is exactly the property we want to investigate.

Testing VIX for Breakout Bands

The first test isolates the effect of VIX on the breakout boundaries. Position sizing remains based on historical volatility across all three models.

We compare the following models:

- **Base Trend:** historical-volatility for bands and for sizing.
- **Trend w/ VIX yClose:** adjusted VIX $close_{t-1}$ for bands, historical-volatility for sizing.
- **Trend w/ VIX Open:** adjusted VIX $open_t$ for bands, historical-volatility for sizing.

VIX for Breakout Bands

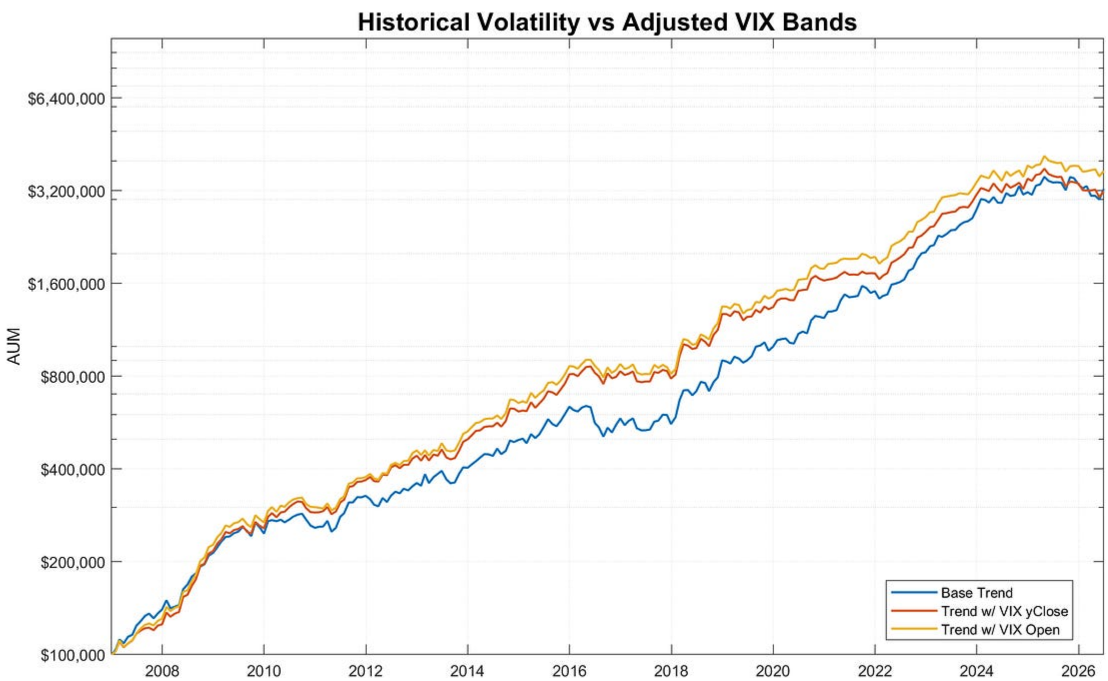
Model	CAGR	Vol	Sharpe	MDD	Daily Hit	Trade Hit	Trades
Base Trend	19.55%	15.67%	1.22	26.61%	45.14%	37.00%	8694
Trend w/ VIX yClose	19.45%	14.87%	1.27	21.21%	45.79%	37.39%	8532
Trend w/ VIX Open	20.38%	14.82%	1.33	19.45%	46.09%	37.55%	8474

The first result is encouraging: using VIX can improve the calibration of the breakout bands.

Even relying on yesterday's VIX close leads to a better risk-adjusted profile than the historical-volatility base model. The **Sharpe ratio improves from 1.22 to 1.27,** while CAGR deteriorates only marginally, from 19.55% to 19.45%. At the same time, realized volatility and maximum drawdown both decline, suggesting that implied volatility already adds useful information even with a one-day lag.

A further source of improvement comes from using **VIX at the open of the same trading day**. In this case, the **Sharpe ratio rises to 1.33 and CAGR improves to 20.38%, while maximum drawdown falls to 19.45%**.

This result suggests that we should **avoid introducing unnecessary lag in the VIX data.**



Equity lines for the base model and the two adjusted-VIX boundary variants.

Testing VIX for Position Sizing

The next question is whether VIX can add value not only when calibrating the breakout signal, but also when sizing the trades.

This distinction matters. Bands determine when we trade. Sizing determines how much risk we take once a trade is active.

If implied volatility is better at capturing the risk environment at the start of the day, it may help the strategy calibrate exposure more effectively and target a more stable level of risk over time.

For this test, we start from the improved ***Trend w/ VIX Open*** model above and then replace historical-volatility-based position sizing with VIX-based position sizing.

VIX for Bands and Position Sizing

Model	CAGR	Vol	Sharpe	MDD	Daily Hit	Trade Hit	Trades
Base Trend	19.55%	15.67%	1.22	26.61%	45.14%	37.00%	8694
Trend w/ VIX yClose	19.45%	14.87%	1.27	21.21%	45.79%	37.39%	8532
Trend w/ VIX Open	20.38%	14.82%	1.33	19.45%	46.09%	37.55%	8474
Trend w/ VIX Open + VIX Sizing	20.53%	14.46%	1.37	17.24%	46.09%	37.55%	8474

This is another constructive result.

Using adjusted VIX at open for the bands already improves the Sharpe ratio from 1.22 to 1.33. When we also use adjusted VIX at open for position sizing, **the Sharpe ratio increases further to 1.37.**

Does VIX Help on Macro Event Days?

The last test focuses on a narrower hypothesis.

If historical volatility is weakest when the market already knows a major catalyst is coming, then implied volatility should be most useful around important macro events.

To test this, we use a **macro-event database** sourced from the **Investing.com economic calendar**.

We focus on the following releases:

- CPI
- FOMC Meeting
- ISM Manufacturing PMI
- ISM Non-Manufacturing PMI
- Initial Jobless Claims
- Nonfarm Payrolls

For this test, **Base Trend** is the standard historical-volatility strategy. **Trend w/ VIX Open + VIX Sizing** uses adjusted VIX open both to calibrate the breakout bands and to size positions. We then condition the daily returns of both strategies on macro-event and non-event days.

Performance Conditional on Macro Events

Event	Strategy	Days	Trigger %	Ret (bps)	Hit Ratio	IR
CPI	Base Trend	231	70.56%	-2.18	42.59%	-0.37
CPI	Trend w/ VIX Open + VIX Sizing	231	69.70%	1.71	45.00%	0.35
FOMC	Base Trend	146	62.33%	4.01	38.46%	0.56
FOMC	Trend w/ VIX Open + VIX Sizing	146	56.85%	3.52	36.14%	0.49
ISM Mfg	Base Trend	228	71.93%	9.14	49.39%	1.25
ISM Mfg	Trend w/ VIX Open + VIX Sizing	228	71.93%	8.38	48.17%	1.30
ISM Non-Mfg	Base Trend	228	67.54%	7.40	45.10%	1.14
ISM Non-Mfg	Trend w/ VIX Open + VIX Sizing	228	67.11%	4.76	45.10%	0.83
Claims	Base Trend	959	68.51%	8.64	46.72%	1.31
Claims	Trend w/ VIX Open + VIX Sizing	959	66.63%	10.66	49.29%	1.83
NFP	Base Trend	226	69.03%	10.96	50.64%	1.63
NFP	Trend w/ VIX Open + VIX Sizing	226	69.03%	10.87	48.72%	1.67
Event	Base Trend	1789	69.09%	6.98	45.74%	1.06
Event	Trend w/ VIX Open + VIX Sizing	1789	66.91%	8.76	46.69%	1.46
No Event	Base Trend	3114	60.73%	7.93	44.75%	1.32
No Event	Trend w/ VIX Open + VIX Sizing	3114	60.85%	7.31	45.71%	1.31

The macro-event results are constructive, but not uniformly so.

Across all selected event days, the full VIX model improves average return from 6.98 bps to 8.76 bps per day, while **improving the information ratio from 1.06 to 1.46.** On non-event days, the result is much less compelling: average return declines from 7.93 bps to 7.31 bps, while IR is essentially unchanged at 1.31.

At the individual-event level, the improvement is strongest around CPI and Initial Jobless Claims. ISM Manufacturing PMI and NFP show smaller IR improvements, while FOMC Meeting Minutes and ISM Non-Manufacturing PMI deteriorate. This suggests that implied volatility can help around scheduled catalysts, but the effect is not uniform across all macro categories.

Conclusion

The main takeaway from this study is that **VIX can add value to an intraday trend-following model, but the way we use it matters.**

Starting from our historical-volatility base model, the strategy delivers a CAGR of 19.55% with a Sharpe ratio of 1.22 over the 2007-2026 sample. Replacing historical volatility with adjusted VIX at the open to calibrate the breakout bands **improves the Sharpe ratio to 1.33**, while also **reducing maximum drawdown from 26.61% to 19.45%.**

The best result in this study is obtained when adjusted VIX at the open is used both to calibrate the breakout bands and to size positions. This full VIX-enhanced model **improves CAGR from 19.55% to 20.53% and Sharpe ratio from 1.22 to 1.37**, while **reducing maximum drawdown from 26.61% to 17.24%.**

This is a meaningful improvement, especially because it is achieved **without changing the underlying trading logic.** What changes is the volatility input used to define the risk environment before the trading session begins.

The macro-event analysis is more nuanced. On the aggregate event bucket, the full VIX-enhanced model **improves average daily return from 6.98 bps to 8.76 bps and information ratio from 1.06 to 1.46.**

However, this improvement is **not homogeneous across event categories.**

In fact, the strongest contribution appears to come mainly from **CPI and Initial Jobless Claims days.** For the other macro events, the improvement is either much smaller, broadly unchanged, or even negative. This means we should be careful not to overstate the macro-event result. VIX seems to help around some scheduled catalysts, but **this part of the research still requires a more dedicated follow-up study.**

A natural extension would be to build a more selective macro-event framework, possibly separating events by release time, surprise magnitude, and volatility regime. Another important direction is to test whether shorter-dated implied-volatility indexes, such as **VIX9D and VIX1D**, can provide a cleaner and more responsive signal than standard VIX for intraday trading applications.

For now, the evidence suggests that **implied volatility can be a useful variable for better calibrating and identifying risk in an intraday trend model.**

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